CS231P - Parallel and Distributed Computing

Group 20: Homework 1

Analysing performance of a shared-memory
distributed computing system under assumed workloads

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Simulation Results

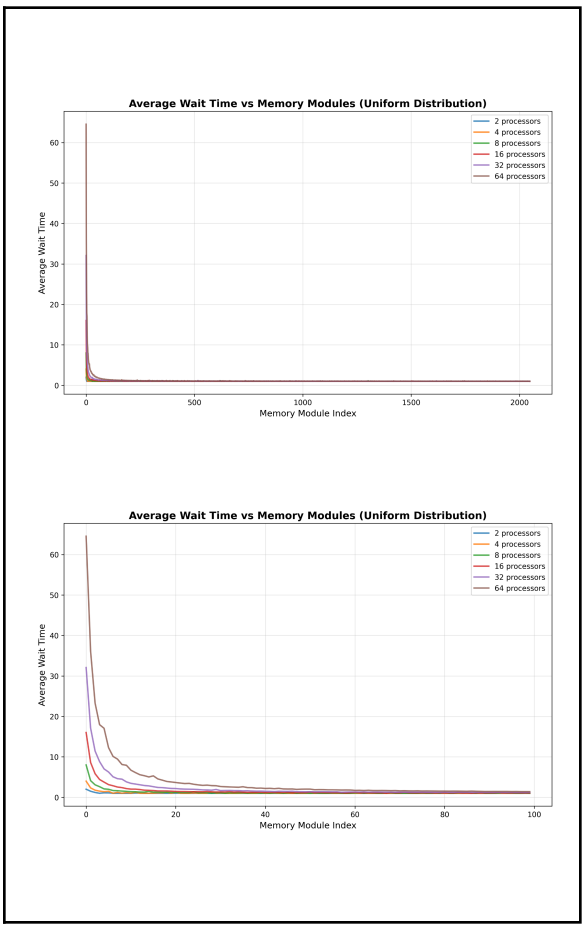

Figure 1: Uniform Distribution
W vs. Memory Modules (m = 1 to 2048)

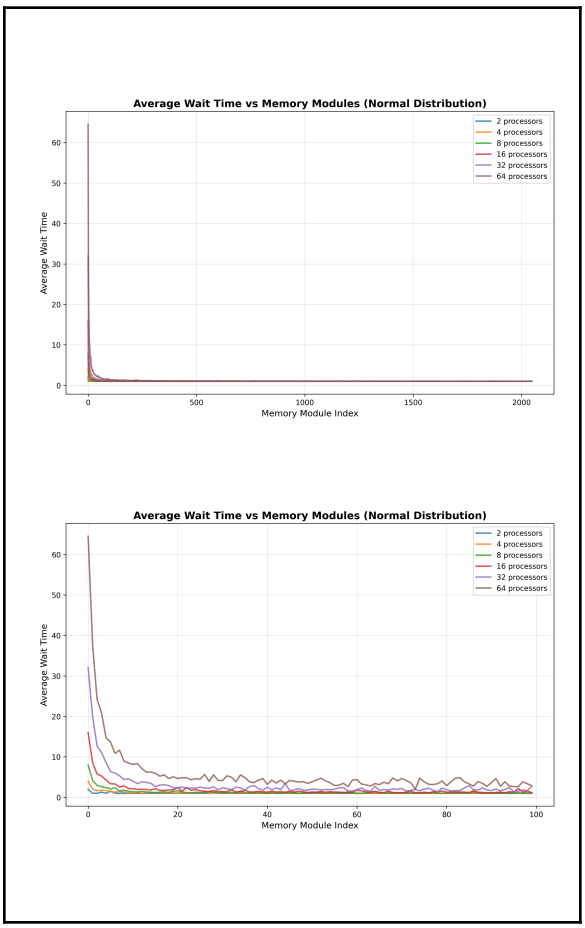

Figure 2: Normal Distribution
W vs. Memory Modules (m = 1 to 2048)

Discussion

1. Effect of Memory Request Distribution

Under a **uniform distribution**, every processor selects a memory module entirely at random, spreading requests evenly across all M modules. This results in contention that is purely a function of the ratio p/m: when m is small relative to p, many processors collide on the same modules; as m grows, collisions become rare and W approaches 1.0.

Under a **normal distribution**, each processor has a fixed preferred module drawn at initialization. Requests cluster around that mean with a narrow standard deviation. This creates hotspot modules that attract disproportionate traffic. Even when the total number of modules m is large, a subset of modules near popular means will experience elevated contention, while the majority of modules remain idle. The practical consequence is that W under the normal distribution may remain elevated for certain processor counts even at large m.

Under a uniform distribution, the wait times decay smoothly and stabilize, as spreading requests evenly across all modules effectively eliminates bottlenecks. In contrast, the normal distribution produces significantly more volatile and elevated latency, particularly visible in the p = 64 curve. Even at m = 64, the wait time remains higher than the uniform baseline, suggesting that the concentration of requests toward specific modules creates persistent "hot spots" that additional capacity cannot fully resolve.

2. Recommendation for Memory Module Procurement

If this simulation informs a hardware procurement decision, specifically how many memory modules to purchase for a system with p powerful processors, the following observations guide the recommendation:

Diminishing returns threshold: The curves in both charts exhibit a knee beyond which W flattens toward 1.0. Adding memory modules beyond this knee yields negligible performance improvement. The knee typically appears around m equal to p for uniform workloads and may require m greater than p for normal workloads due to hotspots.

If the actual workload exhibits strong locality, purchasing many modules beyond the hotspot threshold wastes the budget. Conversely, a uniform workload benefits proportionally from additional modules up to the knee.

Processor Count (p)	Recommended Modules (m) For uniform distribution	Recommended Modules (m) For normal distribution
2	2	8
4	5	15
8	16	25
16	24	50
32	32	80
64	64	100+

3. Conclusion

The simulation demonstrates that the relationship between memory modules and average access time is non-linear and workload-dependent. A uniform distribution produces smooth, predictable contention reduction as m grows, while a normal distribution with narrow variance introduces persistent hotspots. For procurement decisions, the recommended strategy is to identify the knee of the W curve under the anticipated workload distribution and acquire modules up to that threshold, balancing cost against diminishing performance gains.